

Explainable AI for Multiple Sclerosis Detection in MRI FLAIR Images

From CNN classification to post-hoc interpretability (Grad-CAM, Integrated Gradients, LIME)

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Deep Learning achieves strong performance in medical imaging, but clinical adoption is limited by the **black-box** nature of neural networks.

- In healthcare, we need not only *what* the model predicts, but *why*.
- Explainability can increase trust, support debugging, and reveal spurious shortcuts.
- Multiple Sclerosis (MS) in FLAIR MRI is a relevant case: lesions can be subtle, small, and heterogeneous.

Goal of this project: prioritize interpretability over pure accuracy.

Project Summary (What we do)

We study the applicability of Explainable AI (XAI) techniques for MS identification from MRI FLAIR 2D axial slices. The objective is **not** to reveal the internal reasoning of a CNN.

Key components

- Dataset: MSLesSeg (FLAIR, axial 2D slices)
- Framework: PyTorch + Captum + LIME
- Method: supervised learning + post-hoc interpretability analysis

Dataset: MSLesSeg (Axial FLAIR)

- 2D slices extracted from 3D MRI volumes (FLAIR sequence)
- Two groups: **MRIm**s_kde (with lesions), **MRIcon**trol_kde (controls)
- Official split is patient-level (train/test)
- Ground-truth masks available (`_gt.png`) for lesion presence

Note: only the axial view is used, as required.

Data Refinement & Preprocessing

The data pipeline evolved across three dataset iterations to improve label clarity, reduce artifacts, and enable mask-based XAI validation:

- 1 **MSLesionDatasetCached**: fast RAM caching; labels from folder (MS vs control); returns (image, label).
- 2 **MSLesionDatasetClean**: intermediate refinement to study slice-level label noise by inspecting `_gt.png` masks (lesion presence). This version was used to understand whether filtering lesion-free MS slices affects learning and XAI.
- 3 **MSLesionDatasetFinal**: consolidated dataset returning (image, label, mask) with synchronized augmentation (image and mask transformed consistently). Masks are used *only* for post-hoc XAI validation (IoU, pointing game), not as model input.

Preprocessing: resizing to 128×218 , skull stripping via OpenCV thresholding, and normalization.

Model Architecture (Lightweight CNN)

We designed a simple CNN to keep the model interpretable and reduce overfitting risk.

- 3 blocks: Conv2D + BatchNorm + ReLU + MaxPool
- 2 fully connected layers
- Dropout (0.4) to regularize
- Input: grayscale MRI slice (128×218)

Rationale: simpler architectures are easier to debug and analyze with saliency methods.

Training is performed on the available device (CUDA when available) using a standard supervised pipeline.

- Optimizer: Adam
- Initial learning rate: 0.001
- Loss: weighted Cross-Entropy (higher weight for MS)
- Batch size: 64 | Epochs: 20

Learning rate scheduling: ReduceLRonPlateau reduces LR by 50% when **validation loss** does not improve (patience = 2).

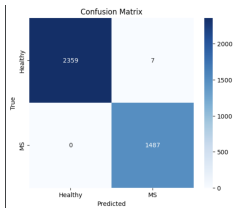
Training Behavior (Loss & Convergence)

- Stable training on GPU (about 15 seconds/epoch)
- Loss decreases from ≈ 0.11 to ≈ 0.069
- Scheduler reduces LR from 10^{-3} to 2.5×10^{-4} , enabling finer convergence

Takeaway: the model converges smoothly without clear signs of instability.

Performance (Confusion Matrix)

Although performance is not the primary goal, it provides context for XAI analysis.



Test summary (slice-level):

- Accuracy = 0.9982 on 3853 slices (≈ 7 errors)
- Precision/Recall are reported as 1.00 in the classification report (rounded to 2 decimals)
- Support: 2366 Healthy (0), 1487 MS (1)

Key point: near-perfect classification does not automatically imply lesion-driven reasoning; we evaluate explanation-mask agreement using IoU and pointing-game tests.

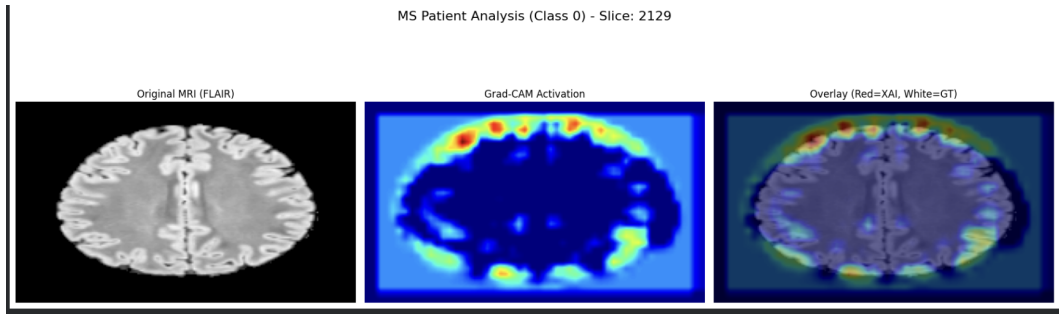
Three complementary XAI approaches were implemented:

- ① **Grad-CAM:** highlights coarse regions important for a class, using the last convolutional layer.
- ② **Integrated Gradients:** pixel-level attribution with axiomatic properties (completeness).
- ③ **LIME:** perturbation-based local explanation using superpixels (model-agnostic).

Focus: verify whether model attention aligns with plausible lesion-related regions or relies on spurious artifacts.

Grad-CAM: Healthy Example (Class 0)

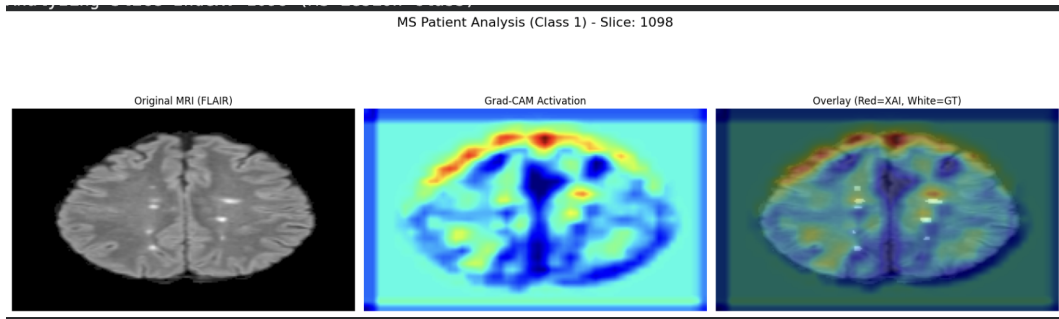
Grad-CAM provides a coarse localization map based on gradients flowing into the last convolutional layer (conv3).



Observation: activation tends to emphasise broad anatomical edges and high-contrast structures rather than focal lesion regions (mask is empty for controls).

Grad-CAM: MS Example (Class 1)

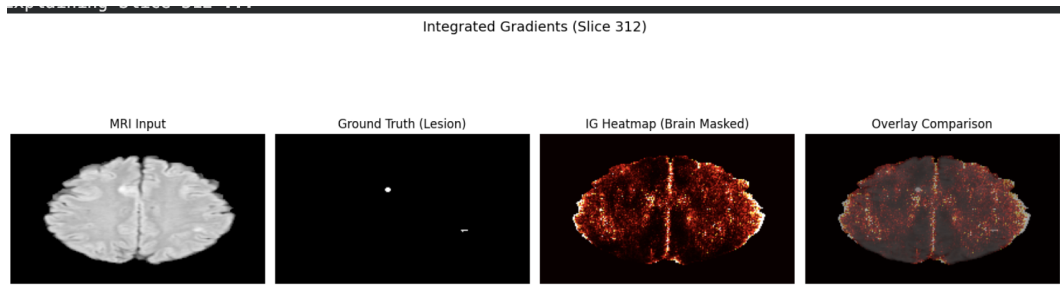
For MS examples, Grad-CAM may still look diffuse because the task is **slice-level classification**, not lesion segmentation.



Interpretation: activation highlights regions influencing the MS prediction; it is not expected to perfectly match lesion boundaries.

Integrated Gradients (SmoothGrad + Brain Masking)

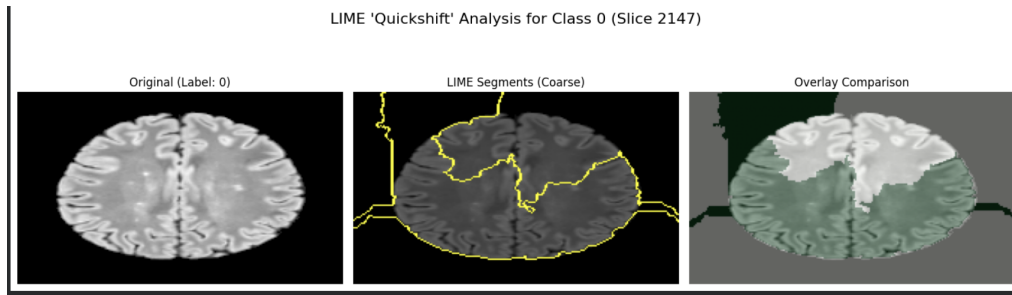
Integrated Gradients provides pixel-level attribution. SmoothGrad and brain masking reduce noise and suppress background artefacts.



Takeaway: smoothing + masking produces a more stable attribution map and improves peak alignment with lesion regions compared to raw IG.

LIME: Class 0 Example (Control / Healthy)

LIME perturbs superpixels and fits a local surrogate model. It is useful as a model-agnostic sanity check.

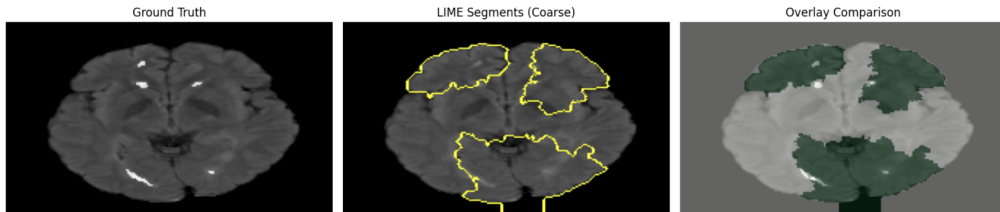


Observation: selected superpixels can be coarse; the main requirement is that explanations remain within meaningful brain tissue rather than background.

LIME: Class 1 Example (MS)

On MS slices, LIME highlights superpixels that increase the probability of the MS class.

LIME 'quickshift' Analysis (Slice 392)



Interpretation: LIME is qualitative and sensitive to segmentation granularity; it should not be interpreted as lesion segmentation.

Takeaway: LIME acts as a sanity check: influential superpixels should correspond to meaningful brain tissue rather than background.

Quantitative XAI Validation (IoU & Pointing Game)

To validate explanations beyond visual inspection, saliency maps are compared with lesion masks using:

- **IoU**: overlap between a thresholded explanation region and the ground-truth lesion mask.
- **Pointing game**: whether the peak attribution falls inside the lesion (strict) or near it (relaxed).

Results (lesion-positive slices, $n = 1487$):

- **Grad-CAM (masked)**: IoU = 0.0197; relaxed hit = 0.0868; strict hit = 0.0034
- **IG (SmoothGrad + masking)**: IoU = 0.0027; relaxed hit = 0.2495; strict hit = 0.2468
- **LIME (20 lesions)**: IoU = 0.0285; intersection hit = 0.55; relaxed hit = 0.05; strict hit = 0.00

Takeaway: explanation-mask agreement is limited overall; smoothing + masking improves IG peak alignment; LIME can show higher overlap on a small subset but remains coarse and unstable.

- The CNN achieves strong test performance (no false negatives; some false positives).
- Captum + LIME enable multiple explanation modalities (gradient-based and perturbation-based).
- Mask-based metrics show that explanations often remain **diffuse** relative to small lesion masks (low IoU).
- **SmoothGrad** + **brain masking** improves Integrated Gradients peak localisation (higher pointing-game hit rate).

Main message: performance alone is insufficient; quantitative XAI validation provides a more rigorous view of whether the model relies on lesion regions.

- **slice-level classification vs lesion-level masks**; XAI maps are not segmentation outputs.
- **Validation split**: created by random slice splitting (not patient-grouped), potentially optimistic.
- **Future work**:
 - patient-level validation split using patient identifiers,
 - perturbation tests (mask lesion region vs random region) to test causal reliance,
 - multi-task learning (classification + lesion segmentation) for stronger lesion-grounded reasoning.

Questions?